

## **ARTHRO-S Powder for Oral Solution**

### **COMPOSITION:**

Each 4.7g sachet contains:

Glucosamine Sulphate Potassium Chloride 1990mg equiv. to Glucosamine Sulphate 1500mg.

Chondroitin Sulphate Sodium 1200mg.

### **DESCRIPTION:**

Appearance of powder:

An orange powder with odour of orange flavour.

Appearance of solution:

An orange solution with odour of orange flavour.

### **PHARMACODYNAMICS:**

Glucosamine is a natural substance found in chitin, mucoproteins, and mucopolysaccharides. It is involved in the manufacture of glycosaminoglycan, which forms cartilage tissue in the body; glucosamine is also present in tendons and ligaments. Glucosamine must be synthesized by the body but the ability to do this decline with age. Glucosamine and its salts have therefore been advocated in the treatment of rheumatic disorders including osteoarthritis.

Glucosamine also acts to improve the viscosity of synovial fluid by increasing synovial fluid production, thereby providing lubricant activity.

Chondroitin sulphate is an important component of most vertebrate tissues. It is most abundant in those tissues with a large extracellular matrix eg. those that form the connective tissues, cartilage, skin, blood vessels, bone, ligaments and tendons. It has anti-inflammatory activity and it affects the cartilage metabolism. Chondroitin sulphate acts as an extracellular proteases inhibitor which is involved in the metabolism of connective tissues. An increase of proteoglycans concentration of the pericellular matrix as well as a dose-dependent decrease in collagenolytic activity is due to the chondroitin sulphate to the medium of a chondrocytes culture from human articular cartilage. Oral administration of chondroitin sulphate has shown to stimulate proteoglycan production of chondrocytes which inhibit the production of certain proteolytic enzymes.

Glucosamine and chondroitin sulphate combination: Glucosamine and chondroitin sulphate synergistically act to stimulate glycosaminoglycan synthesis. Combining these substances produces, from 1 site, the upregulation of the synthetic activities in chondrocytes as well as the inhibition of degradative enzymes from the other site.

Analgesics and NSAIDs may be used concomitantly with this product, either for the prevention of analgesia during possible of the disease, or in the initial period of treatment, when the symptomatic effects of the product may be delayed for few weeks. Physical therapy programs can be used concomitantly with this product in the overall management of osteoarthritis.

### **PHARMACOKINETICS:**

Glucosamine:

#### ***Absorption***

After oral administration, bioavailability is low due to first-pass hepatic metabolism~26%.

The gastrointestinal absorption is close to 90%.

***Distribution***

Glucosamine is not protein-bound, but rather incorporates into plasma proteins (primarily globulins)

Volume of Distribution: 2.5 Liters

***Metabolism***

Liver, extensive

The first-pass effect in the liver in which more than 70% of glucosamine is metabolized.

***Excretion***

Renal Excretion,

10% Feces, 11%

Part of a dose of glucosamine sulfate is eliminated as carbon dioxide via expired air.

Chondroitin sulphate:

***Absorption***

After oral administration, the bioavailability is 15-24%. A 10% of absorbed chondroitin sulphate appears in unmetabolized form and 90% as depolymerized derivatives with lower molecular weight, as a result of first-pass effect metabolism.

***Distribution***

The distribution volume of chondroitin sulphate is relatively low. In humans, it has shown to have affinity for joint tissues. At least 90% of chondroitin sulphate dose is first metabolized by lysosomal sulfatases, and after that it is depolymerized by hyaluronidases ie,  $\beta$ -glucuronidases and  $\beta$ -N-acetyl hexose amidases. Liver, kidneys and other organs are involved in the depolymerization of chondroitin sulphate. Chondroitin sulphate remains in the body for 5-15 hours.

***Excretion***

Main part of chondroitin sulphate and its depolymerized derivatives are eliminated via the kidneys.

**INDICATION:**

Adjunctive therapy for osteoarthritis

**ROUTE OF ADMINISTRATION:**

Oral administration.

**RECOMMENDED DOSE:**

The content of one sachet (dissolved in glass of water) should be taken once daily, preferably at meals.

**CONTRAINDICATIONS:**

Contraindicated in individuals who have hypersensitivity to glucosamine or to any of the excipients. As the active ingredient is obtained from seafood (shellfish), the product should not be given to patients who are allergic to shellfish.

Contraindicated in individuals who have hypersensitivity to chondroitin sulphate.

**WARNING AND PRECAUTION:**

Glucosamine treats the underlying cause of osteoarthritis and the therapeutic effect can only be seen after 2-3 weeks. Therefore, it is advisable to take an analgesic/anti-inflammatory drug if required during the first 2-3 weeks of therapy with glucosamine. Safety and effectiveness have not been

established in children therefore children, should avoid using glucosamine. Patients with severe hepatic or renal insufficiency should be monitored. Derived from seafood, therefore should not be given to patients who are allergic to shellfish. A doctor should be consulted in order to exclude the presence of other joint conditions/diseases for which an alternative treatment should be considered.

#### **EFFECTS ON ABILITY TO DRIVE AND USE MACHINES:**

No effects on the ability to drive or to operate machines are expected.

#### **INTERACTIONS WITH OTHER MEDICAMENTS:**

Effects on glucose metabolism & antidiabetic agents:

It has been hypothesized that glucosamine may impair insulin secretion through competitive inhibition of glucokinase in pancreatic beta cells and/or alteration of peripheral glucose uptake.

Glucosamine may increase insulin resistance and consequently affect glucose tolerance.

It may reduce antidiabetic agent effectiveness eg. when used with these antidiabetic agents:

Acarbose, Acetohexamide, Chlorpropamide, Glipizide, Glyburide, Metformin, Miglitol, Pioglitazone, Repaglinide, Rosiglitazone, Glimepiride, Tolbutamide, Troglitazone.

Glucosamine is likely safe in patients with well-controlled diabetes (HbA1c less than 6.5%) taking one or two oral antidiabetic medications or controlled by diet only. In patients with higher HbA1c levels or those taking insulin, monitor blood glucose levels closely/more frequently.

Reduced effectiveness when used with glucosamine: Doxorubicin, Etoposide, Teniposide  
Warfarin

-Elevations of International Normalized Ratio serum values and potentiation of anticoagulant effects

-If concomitant therapy is necessary, the patient's INR should be more closely monitored.

#### **PREGNANCY AND LACTATION:**

Available evidence is inconclusive or inadequate for use in pregnant or lactating mothers. Until more information is available, this product should only be used under medical supervision in pregnancy and lactating mothers if the potential benefit to the mother justifies the potential risk to the fetus. Administration during the first 3 months of pregnancy must be avoided.

#### **SIDE EFFECTS:**

*Cardiovascular:*

Peripheral oedema, tachycardia were reported in a few patients following larger clinical trials investigating oral administration in osteoarthritis. Causal relationship has not been established.

*Central nervous system:*

Drowsiness, headache, insomnia have been observed rarely during therapy (less than 1%).

*Gastrointestinal:*

Nausea, vomiting, diarrhoea, dyspepsia or epigastric pain, constipation, heartburn and anorexia have been described rarely during oral therapy with glucosamine.

*Skin:*

Skin reactions such as erythema and pruritus have been reported with therapeutic administration of glucosamine.

#### **OVERDOSAGE:**

No cases of accidental or intentional overdose are known. The animal acute and chronic toxicological studies indicate that toxic effects and symptoms of toxicity are not likely to occur, even after high overdoses.

**STORAGE:**

Store in a dry place below 30 °C.

**PRESENTATION:**

Box of 10, 30, 60, 100 or 500 sachets.

**MANUFACTURER AND PRODUCT REGISTRATION HOLDER:**

NORIPHARMA SDN. BHD. 200701034604 (792633-A)

Lot 5030, Jalan Teratai,  
5 ½ Mile off Jalan Meru,  
41050 Klang, Selangor, Malaysia

**DATE OF REVISION:**

April 2024